

MP-H1

A high-sensitivity two-way with a 3D-printed tractrix horn.

An OJAS-inspired two-way: a 12" reflex bass bin and a 1.4" compression driver on a printed round tractrix horn. Minimal passive parts, high-efficiency drivers, and a single amplifier channel with DSP doing the voicing. Everything here — the acoustic maths, the crossover, the print files, the tuning method — is shared so you can build, adapt, or improve on it.



The finished pair — 12" reflex bins and printed tractrix horns, fresh off the bench.

FORMAT

Two-way, high-efficiency
passive XO + DSP voicing

WOOFER

12" Peerless FSL-1225
96.2 dB · reflex

HORN / CD

1.4" SB Audience ROSSO
108 dB · titanium

ENCLOSURE

62 L reflex · $f_b \approx 47$ Hz
 $F_6 = 47$ Hz

CROSSOVER

≈ 1.2 kHz · 2nd-order
on the CD only

AMPLIFICATION

Single channel + DSP
NAD C 700 · Dirac Live

01 The idea

MP-H1 · TWO-WAY HORN LOUDSPEAKER · BUILD & DESIGN NOTES

CONCEPT & PHILOSOPHY
TOM ANDREWS · OPEN BUILD

A modern take on the OJAS / Devon Turnbull two-way: horn-loaded efficiency and effortless dynamics, but designed around drivers you can actually buy and voiced with DSP instead of a wall of passive components.

The MP-H1 pairs a 12" pro woofer in a modest reflex box with a 1.4" compression driver on a 3D-printed round tractrix horn that sits on top of the cabinet. The two drivers share **one amplifier channel**. The design leans on three deliberate ideas:

Minimal passive parts

The woofer runs **direct** — no series inductor, nothing in its path. Only the compression driver carries a passive network (a second-order high-pass and a level-matching L-pad). Four components per speaker, total.

Efficiency first

96 dB woofer, 108 dB horn. The whole system is happy on a handful of watts, so amplifier quality matters more than amplifier power, and the horn's dynamics come for free.

DSP does the voicing

Baffle-step, the low-shelf lift, the 40 Hz excursion high-pass and room correction all live in DSP — not in the crossover. The passive parts only set the acoustic crossover and match levels; tone is dialled after measurement.

The reference is the OJAS two-way (FaitalPRO 15" + compression driver on a long tractrix); the Klipsch×OJAS kO-R2 crosses at 760 Hz with 97 dB sensitivity. The MP-H1 keeps that spirit but re-bases the drivers on parts stocked in Australia, and moves the tone controls into DSP.

02 The drivers

DATASHEET VALUES

Woofer — Peerless by Tymphany FSL-1225R02-08 (8 Ω)

A pressed-steel pro 12" on a paper cone and 65 mm voice coil, spec'd by Tymphany as “extremely flat to 3 kHz.” 96.2 dB sensitivity, Ø313.5 mm frame, ~96 mm deep.

F _S	Q _{TS}	Q _{ES}	V _{AS}	R _E	BL	S _D	X _{MAX}	M _{MS}	L _E	SENS.	POWER
63 Hz	0.44	0.49	54.8 L	5.5 Ω	14.9 Tm	547 cm ²	5.3 mm	50.1 g	0.54 mH	96.2 dB	233 W

Compression driver — SB Audience ROSSO-65CD-T (8 Ω)

- 108 dB (1 W / 1 m), 70 W AES / 140 W max, R_{dc} 6.2 Ω
- Range 500 Hz – 20 kHz; **recommended crossover 1 kHz**
- 65 mm **titanium** diaphragm, ferrite motor, annular phase plug, copper-cap demodulation
- **1.4" throat** (35.6 mm exit), 18.4° conical
- Flange **4 × M6 on Ø102 mm PCD** — the printed horn bolts straight on
- Body Ø156 mm, 65.9 mm deep, **3.76 kg** — carried by the printed stand, not the horn flange

Both drivers are stocked in Australia by Wagner Online (SB Audience / Peerless / Dayton distributor). They replaced the original Celestion pair, which went unobtainable locally — the cabinet and horn were designed to accept either without change.

The enclosure is a **62 L net** reflex box (456 × 536 × 366 mm external, 18 mm Baltic birch) with a full-width front slot port. Its tuning wasn't chosen as a target — it falls out of the box volume and the driver:

The alignment

$\alpha = V_{as}/V_b = 54.8 / 62 = \mathbf{0.88}$
 $h = f_b/F_s = \mathbf{0.76} \rightarrow f_b \approx 0.76 \times 63 = \mathbf{48\ Hz}$
as-built 150 mm duct $\rightarrow \sim 47\ Hz$

Why not lower

The FSL's Q_{ts} of 0.44 is too high for a maximally-flat B4. A gentle **QB3** alignment suits it — and 5.3 mm X_{max} would run out of excursion long before 30 Hz. This is a high-sensitivity monitor, not a sub.

The result is a deliberately shallow roll-off. A DSP low-shelf (+3–4 dB below 100 Hz) plus normal room gain lifts the in-room response flat into the **mid-40s**.

FREQ (HZ)	40	45	50	55	60	70	80	100	150	200
Response (dB)	−9.0	−6.7	−5.3	−4.4	−3.7	−2.7	−2.0	−1.2	−0.4	−0.2

Extension & excursion

F3 = 67 Hz, **F6 = 47 Hz**, F10 = 38 Hz. Cone travel is $\sim 2.5\ mm$ peak at 30 W across 48–200 Hz, well inside the 5.3 mm X_{max} . Below f_b the cone unloads, so a **40 Hz / 24 dB/oct DSP high-pass** protects it — essential once the low-shelf is in.

The port

A full-width slot at the base of the baffle: **420 × 28 mm** ($S_p = 117.6\ cm^2$), 150 mm deep, formed by a shelf above the cabinet floor. Peak air velocity $\sim 16.5\ m/s$ at 30 W — just under the $\sim 17\ m/s$ chuffing threshold at $\sim 112\ dB$ peaks.

04 The horn

MP-H1 · TWO-WAY HORN LOUDSPEAKER · BUILD & DESIGN NOTES

TRACTRIX · 3D PRINTED
TOM ANDREWS · OPEN BUILD

A round **tractrix** flare expands so the wavefront stays roughly spherical as it grows — smooth loading, minimal reflection back down the throat. The whole curve comes from one number, the flare cutoff f_c :

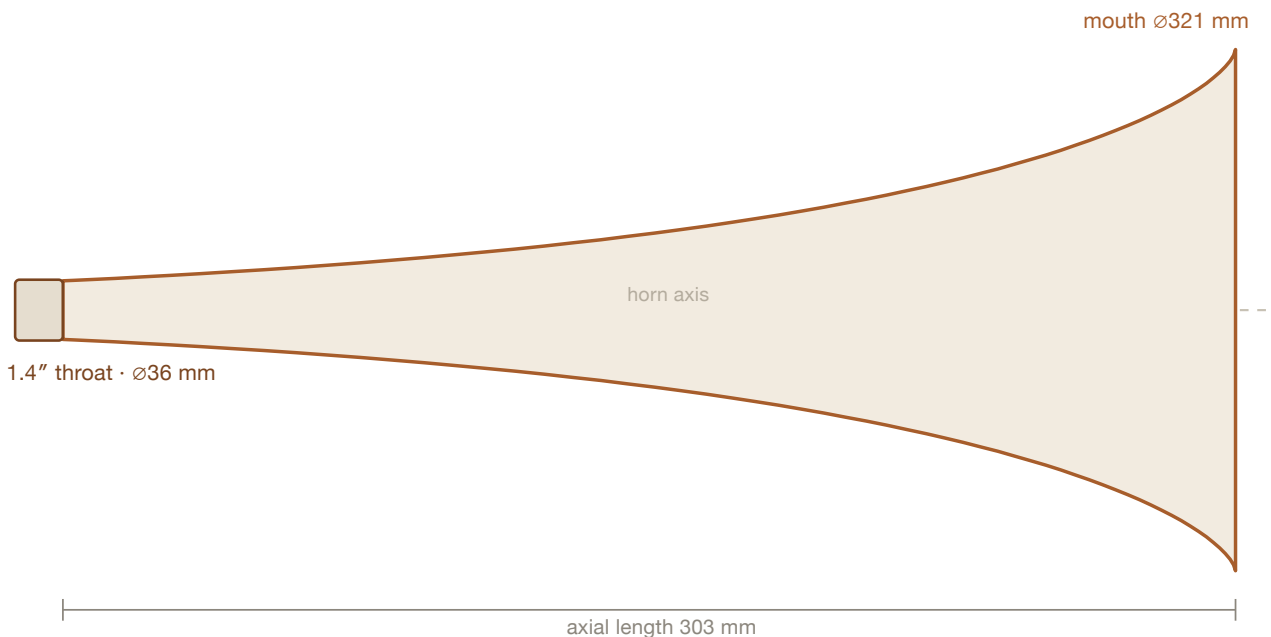
$$r_m = c / (2\pi \cdot f_c) \quad \bullet \quad x(r) = r_m \cdot \ln((r_m + \sqrt{(r_m^2 - r^2)}) / r) - \sqrt{(r_m^2 - r^2)}$$

The numbers

- Throat **Ø36 mm** (fixed by the 1.4" driver)
- $f_c = 340 \text{ Hz} \rightarrow$ mouth **Ø321 mm**, length **303 mm**
- Loads effectively from **~450 Hz** — over 1.5 octaves below the ~1.2 kHz crossover
- 6 mm walls throughout; the print must be *dead* (a dull thock, not a ring)

Printed in sections

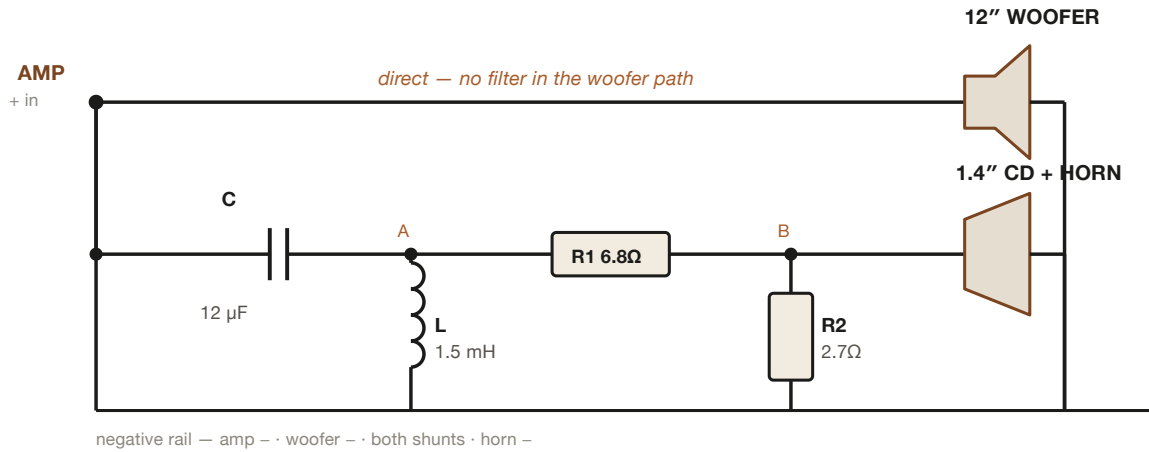
Too big for the bed, so it splits into one **neck** (with the driver bolt flange printed in) and **four identical bell quadrants** — a single STL rotated 90° four times. Every seam carries an **integral tongue-and-groove** printed into the parts: a continuous ring at the neck join, plus three tabs per quadrant seam. No hardware — dry-fit, epoxy in the grooves, tape, cure.



Round tractrix profile — flare cutoff $f_c = 340 \text{ Hz}$. Plotted from the design equation.

The clever bit: tongue on the trailing face, groove on the leading face, so one quadrant self-tiles into a closed ring under rotation. Clearances (0.15 mm width / 0.2 mm depth) are tunable in the generator script for a tighter or looser printer.

Both drivers hang on one amp channel. The woofer is wired direct; the compression driver gets a **2nd-order Butterworth high-pass** then a **level-matching L-pad**.



Signal path, one amp channel. Woofer taps the input directly; the CD chain is C+L (high-pass) then R1/R2 (L-pad).

High-pass — the crossover

12 μF series + 1.5 mH shunt gives $f_c = 1.19 \text{ kHz}$, $Q \approx 0.72$ — a clean second-order Butterworth, safely above the ROSSO's 1 kHz minimum. The woofer needs no low-pass: its own roll-off plus the horn's takeover cover the handover.

L-pad — the level match

The 108 dB horn is dropped $\sim 12 \text{ dB}$ to meet the 96 dB woofer: series 6.8 Ω / shunt 2.7 Ω (10 W wirewound). It also keeps the CD seeing a steady $\sim 8 \Omega$, which damps the horn's impedance peaks so the crossover stays put. Final trim is by ear / REW.

System impedance dips to $\sim 4.4 \Omega$ near crossover — safe on any 4 Ω -rated channel. Terminals are four binding posts: rear pair = amp in, top pair feeds the external horn up through the lid. Start with the compression driver **inverted** and confirm the summed response in REW.

The passive network sets the crossover and matches levels; **everything tonal is DSP**. A calibrated mic (UMIK-1) and REW turn the raw speaker into a voiced one in an afternoon. Enter these baseline blocks first — they're design maths, not taste:

DSP BLOCK	SETTING	WHY
High-pass	40 Hz, 24 dB/oct	protects the woofer below f_b where the box unloads the cone
Low-shelf	+3 dB <100 Hz, Q 0.7	fills the shallow QB3 roll-off / baffle step
Peak (placeholder)	−2 dB @ 1.6 kHz, Q 2	typical woofer/horn overlap bump; refine by measurement

Measurement pass

- **Nearfield:** confirm the port peaks ~47 Hz — verifies box tuning before any EQ.
- **CD polarity:** flip the horn leads, keep whichever sums smoothly through 1–1.5 kHz.
- **Level match:** horn band vs woofer band; if $> \pm 1.5$ dB, change the L-pad *series* resistor, don't burn DSP headroom.

Room & target

- Multi-position sweep at the seat; correct room modes **below 500 Hz** freely.
- Above 500 Hz, only broad trends ($Q \leq 2$, cuts preferred) — never fill narrow dips.
- Target: flat 200 Hz–2 kHz, then a gentle downward tilt (~−4 dB by 10 kHz). Trust your ears for the last dB.

This build uses a NAD C 700 with **Dirac Live** — chosen for impulse-response correction, because the horn's acoustic centre sits well behind the woofer's and that arrival-time error can't be fixed by magnitude EQ alone. Start with the limited-bandwidth licence (corrects to ~500 Hz — enough for room modes and the baffle step).

07 Bill of materials

MP-H1 · TWO-WAY HORN LOUDSPEAKER · BUILD & DESIGN NOTES

PER PAIR · EST. AUD
TOM ANDREWS · OPEN BUILD

GROUP	KEY ITEMS	EST. AUD
Drivers	2× FSL-1225 12" woofer, 2× ROSSO-65CD-T compression driver	850–1,000
Crossover	2× 12 µF MKP cap, 2× 1.5 mH air-core coil, 4× wirewound resistor	~100
Timber & hardware	18 mm Baltic birch, feet, binding posts, T-nuts, wadding, wire	~440
Horn & finish	~5 kg PETG, stand hardware, filler, primer, topcoat	~255
Total	excludes the measurement mic (~\$150 if not owned)	1,650–1,850

Prices checked mid-2026; small-run pro-audio stock moves — confirm at order time. Most of the electronics ship in one box from a single distributor; plywood and timber are the only local buys.

08 Field notes

WHAT THE BUILD TAUGHT US

The one that nearly fooled us — check polarity first

First power-up: the bass was *gone* — thin and lifeless. The tempting move is to blame the port tuning or the crossover. Instead: an AA battery across each woofer's terminals, watching the cones. One punched **out**, the other pulled **in** — one woofer was wired out of phase, and the two were cancelling each other acoustically in the room. Two minutes to fix. They went from lifeless to *alive*. **Battery-test your polarity before you second-guess the acoustics.**

- **Cut the woofer hole to the driver's real cutout**, not the frame OD — the flange must land on solid baffle. Cut slightly small, open out on a test-fit.
 - **Seal everything.** A reflex box that leaks loses bass — bed the top posts and gasket the woofer.
- **Register your printed seams.** The quadrant-to-quadrant joints, not the neck join, are where misalignment shows — give them positive location.
 - **Leave the L-pad resistors swappable** (long leads / a lever-nut) — you re-trim by ear once you measure *this* horn.

Designed & built by Tom Andrews on the Mornington Peninsula. Acoustic maths from `box_model.py` / `tractrix.py`; horn generated headless in `build123d`. Round tractrix, single-amp, high-sensitivity two-way.

NEXT REVISION
MP-H2
elliptical R-OSSE waveguide